PASCAL Muhire

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A. PERSONAL DETAILS

Date and place of Birth: Rusizi District, on July 08, 1997

Nationality: Rwandan
Marital Status: Single
Residence address: KN 22 St, Nyarugenge, Kigali

B. PROFILE

I am a passionate advocate for sustainable resource management, with a strong commitment to making a positive impact on society through the development of mechanical and energy infrastructure and environmental stewardship. My professional journey spans roles in engineering design, resource assessment and analysis, low-carbon energy design, modelling, and feasibility studies.

At CIMERWA, formerly Prime Cement, I served as **a Mechanical Engineer**, contributing to the construction and maintenance of Silos and other parts of the cement plant. My responsibilities included the construction of silos, feeders and other parts of the plant and their maintenance.

As a Professional Intern at Rwanda Standard Board (RSB), I contributed to designing and implementing audits in different industries that deal with mechanical products, preparing technical drawings and reports, ensuring environmental compliance, and overseeing all activities to meet quality standards, providing feedback to staff and management to promote quality products that are sustainably aligned with local needs and regulatory requirements.

During my **Professional internship at Rwanda Energy Group (REG)**, I gained hands-on experience in the supervision and maintenance of the electricity Transmission infrastructure. I also participated in energy planning, where programs like the Rwanda Universal Energy Access Program (RUEAP) and ASCENT. This multidisciplinary experience has equipped me with technical, analytical, and project management skills essential for effective energy resource and environmental engineering.

C. KEY SKILLS

- **Engineering Design Projects:** Experience in engineering design projects, problem-solving, prototyping, and technical development**.**
- **Machine Design and CAD:** Ability to design mechanical and Energy systems components using engineering drawing, CAD, computer-aided manufacturing, and machine element design.
- **Mechanical Vibrations and Dynamics:** Ability to analyse vibration, dynamic behaviour of machines, and mechanical system stability.
- **Control Systems Engineering:** Understanding of control systems, instrumentation, measurements, and automation in mechanical and industrial systems.
- **Advanced Mechanical and Energy Materials:** Understanding of materials used in Mechanical design and energy applications, including photovoltaic, storage, and thermal systems.
- **Thermal Engineering**: Application of thermodynamics, heat transfer, and thermal sciences in energy conversion and efficiency improvement.
- **Experimental Design and Data Analysis:** Ability to conduct laboratory experiments, analyze engineering data, and interpret results for research and system optimization.
- **Applied Computation and Simulation:** Use of computational methods, numerical modelling, and engineering software for simulation and analysis of Mechanical and energy systems.
- **Project Design and Management:** Planning, design, and management of engineering and energy projects from concept to implementation.
- **Energy Economics and Policy:** Skills in economic analysis of energy projects, cost evaluation, techno-economic assessment, and understanding the policy-engineering nexus in the energy sector.

- **Renewable Energy Technologies:** Competence in solar PV, solar thermal, bioenergy, and hybrid renewable systems for sustainable power generation
- **Power and Hybrid Energy Systems Design:** Design and analysis of integrated energy systems combining multiple generation sources for reliable and sustainable supply.
- **Research, Academic Writing, and Presentation:** Strong ability in technical writing, scientific reporting, and professional presentations.
- **Communication and Management Skills:** Ability to communicate technical information effectively and manage multidisciplinary projects.

D. EDUCATION

INSTITUTION AND LOCATION	DEGREE/ CERTIFICATE	YEAR(s)	FIELD OF STUDY
University of Tlemcen (Pan African University, Institute of Water and Energy Sciences including Climate Change) (PAUWES), Algeria	MSc	2024-2026	Master's Degree in Mechanical and Energy Engineering
University of Rwanda, College of Science and Technology (CST), Rwanda	BSc	2018-2022	Bachelor of Science in Mechanical Engineering option of Production Engineering
Lycee de Kigali	A2	2015-2017	Physics, Chemistry and Mathematics (PCM)

E. LANGUAGES

Language	Speaking	Reading	Writing
English	Excellent	Excellent	Excellent
French	Good	Good	Good
Kiswahili	Little	Good	Good
Kinyarwanda	Excellent	Excellent	Excellent (Mother tongue)

F. EXPERIENCE

Date	Job Title	Employer	City, Country
Aug, 2025 – Jan, 2026	Professional Intern	Rwanda Energy Group (REG)	Kigali, Rwanda
July, 2023 – Feb, 2024	Professional Intern (Auditor in the Mechanical field)	Rwanda Standard Board	Kigali, Rwanda
Jan, 2022 – April,2022	Mechanical Engineer Internee	PRIME CEMENT Ltd.	Musanze District, Rwanda

G. RESEARCH TITLE

1. Study of Power-to-Hydrogen -to-Power Concept for Sustainable Power Generation in Rwanda (01 August 2025 – 09 February 2026)

- ⇨ Research project funded by African Union (AU)
- ⇨ Position: Researcher
- ⇨ Country: Rwanda
- ⇨ Summary: This research provides critical insights for planners and the energy sector as a whole, helping them understand what it takes to integrate green hydrogen into the country's energy mix and the technical, economic, and environmental implications.

2. Design of Solar-Powered Food Waste Recycling Machine, Rwanda (17 June 2021 – 30 January 2023)

- ⇨ Research project funded by the University of Rwanda (UR)
- ⇨ Position: Researcher
- ⇨ Country: Rwanda
- ⇨ Summary: The research aimed to contribute to food waste management by designing a solar-powered machine that converts this waste into useful manure. This idea aims to tackle the issues of waste, especially in Kigali.

H. SHORT TRAINING & LEADERSHIP ROLES

Dates		**Course Title and Training Description**	**Award (Give details)**	**Awarding Body**
From	**To**			
27 Jul, 2024	31 Jul,2024	Summer School on Green Hydrogen	Certificate	Ta'ziz project, German
27 Nov, 2024	10 March, 2025	AutoCad	No Certificate	Private Tutor,Kenya
5 Aug,2024	15 Aug,2024	Carbon credits buying and Selling	Certificate	Atingi, Online
Leadership and Management				
2021	2022	Director Of Operations at AUSP	No Certificate	African University Student Platform (AUSP)
2016	2017	President of Hope for the Needy in Lycee de Kigali	Certificate	Hope for the Needy

I. COMPUTER LITERACY

Engineering Softwares and Models

- **AutoCAD**
- **Solidworks**
- **PVSyst**
- **Homer**

Computer Programming

- **C++**
- **MATLAB**
- **Python**

Microsoft Office

- **Word, Excel, PowerPoint, Access, and Publisher**
- **Professionally use the Internet**

J. PROFESSIONAL AFFILIATIONS

1. Member of East Africa Solar Energy Network
2. Member of Solar Energy Professionals - PV & Thermal
3. Member of YOUNGO, the official youth constituency of the UNFCCC

K. REFEREES

S/N	Name	Affiliation
1	Prof. Abdallah Khellaf	Directeur de recherche / CDER, Alger, Algérie. Cell: +213772542525 Email: khellaf.at@gmail.com
2	MURINDI JEAN BOSCO	Division Manager at Rwanda Standard Board Kigali, Rwanda Cell: +250 728 308 530 Email : jb.mulindi@rsb.gov.rw
3	NSHIMIYIMANA JEAN CAPISTRAN	Recruitment and Training specialist at REG/EDCL , Kigali, Rwanda Cell: +250 783874565 Email : canshimiyimana@edcl.reg.rw

L. CERTIFICATION

I, the undersigned, certify that to the best of my knowledge and belief, these data correctly describe me, my qualifications, and experience.

Pascal MUHIRE